

IFT 222: COMPUTER ARCHITECTURE AND ORGANISATION

CSC227 TEST 1

TIME: 45 MINS

ATTEMPT ALL

1. Compare Von Neumann Architecture and Harvard Architecture.
Include diagram for each. 3mks
2. Represent -452.25 as a floating-Point number using the Single Precision IEEE format.
Leave your answer in hexadecimal form. 5mks
3. What do you understand by Overflow and Underflow in Data Representation? 3mks
4. Describe end-around carry in 1's complement. Illustrate this using 125-123 for 8-bit register. 4mks
5. Express 11110111101111001110 in Octal and Hexadecimal form 3mks
6. Explain why it is better to have new type of memory that is based on more transfer rate than more memory size. 2mks

Start: - 4:30
Stop: 6:30
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B.Sc DEGREE EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

SESSION: 2023/2024

COURSE CODE: CSC227

COURSE TITLE: COMPUTER ARCH AND ORGANIZATION I

DEPARTMENT: CIS

SEMESTER: OMEGA

CREDIT UNIT: 2

TIME: 2 HOURS

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

Question One (30 marks)

- a. A 5-stage pipelined processor has Instruction fetch (IF), Instruction decode (ID), Operand Fetch (OF), Execute (EX) and Write Back (WB) stages. The IF, ID, OF and WB stages take 1 clock cycle each for any instruction. The EX stage takes 1 cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction and 6 clock cycles for DIV instruction respectively. Draw the cycle diagram and determine the number of cycles needed to execute the following sequence of instructions
 - i. With Operand forwarding 4½mks
 - ii. Without Operand forwarding 5½mks

I1	MUL R2, R0, R1	; R2 ← R0 X R1
I2	DIV R5, R3, R4	; R5 ← R3/R4
I3	ADD R2, R5, R2	; R2 ← R5 + R2
I4	SUB R5, R2, R6	; R5 ← R2 - R6
- b.
 - i) State Amdahl's law' 2½mks
 - ii) Assume in hypothetical system cache is 10 times faster than main memory and CPU accesses the cache 85% of the time. Using Amdahl's law, calculate the overall speed up of the system. 2mks
- c. Consider three (3) floating point numbers X, Y and Z stored in registers based on the IEEE 754 Single precision floating point format.
 $X = C1400000h$, $Y = 42100000h$, $Z = 41400000h$
Convert X, Y and Z to decimal numbers and determine which of the following is true or false 9mks
 - i. $(Y-Z) > 0$
 - ii. $X + Z = 0$
 - iii. $Z = X + Y$

- d. If an 8-way set association cache is made up of 32 bits words, 4 words per line and 4096 sets.
- i) Size of each line of the cache 1mk
 - ii) Size of each set of the cache 1mk
 - iii) Size of the cache itself 1mk
 - iv) Sketch the structure of the cache 1½mks
- e. Describe how cache memory improves the performance of a computer system 2mks

Question Two (20 marks)

- a. What is Boolean Algebra? 2mks
- b. Design a code converter with two inputs (X_1 and X_0) and four outputs (Y_3 , Y_2 , Y_1 , and Y_0). The truth table is presented in Table 1.

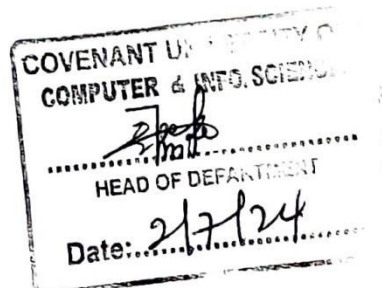
Table 1: Truth Table

X_1	X_0	Y_3	Y_2	Y_1	Y_0
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1

- i). Derive the simplified Boolean Expression for each Output using 2-input k-map. 4mks
 - ii). Draw the full logic circuit of the converter using basic logic gates. Label all inputs and outputs. 4mks
- c. Design a combinational circuit with three inputs X,Y, and Z, and the three outputs, A,B, and C. When the binary input is 0,1,2 or 3, the binary output is one greater than the input. When the binary input is 4,5,6, or 7, the binary output is one less than the input.
- i. Derive the truth table for the given problem 2mks
 - ii. Obtain simplified Boolean expressions using K-map 4½mks
 - iii. Draw the corresponding logic expression 3½mks

Question Three (20 marks)

- a. Determine the average memory access time for a machine with a cache hit rate of 80% and cache access time of 5 ns and main memory access time of 100 ns when.
 - i. Simultaneous access memory organization is used. 2½mks
 - ii. Hierarchical access memory organization is used. 2½mks
 - iii. Draw the diagrams representing the two scenarios 3mks
- b. A computer system uses 16-bit memory addresses. It has a 2K-byte cache organized in a direct-mapped manner with 64 bytes per cache block. Assume that the size of each memory word is 1 byte. Calculate the number of bits in each of the **Tag, Index, and Offset fields** of the memory address. 3mks
- c. Explain the following 4½mks
 - i) Locality of reference



- ii) Temporal locality
 iii) Spatial Locality
 d. Simplify the Boolean expression with Boolean algebra laws and show that 4½ mks

$$(AB'. (A+C))' + A'B. (A+B'+C')' = A'+B$$

Question Four (20 marks)

- a. i) What do you understand by Von Neumann Concept? 1mk
 ii) Mention any 2 significant aspects of Von Neumann Architecture 2mks
 iii) Describe the Von Neumann architecture bottleneck 2mks
 b. i) Complete the table below. Show all necessary work done to arrive at the answer. 6mks
 Use 12bits.

Value	Unsigned	S-M	1's Complement	2's Complement
204				
-139				
1101 00110 11				

- ii) Calculate the memory requirement of an image whose dimension is 17.12 inches by 9.45 inches at resolution 250 dpi. Express your answer in Megabyte 3mks
 c. Represent the following -1240.56 using Single Precision IEEE format for floating point numbers. 6mks

Question Five (20 marks)

- a. Consider the combinational circuit shown in Figure 1
 i. Derive the Boolean expression for T1 through T4 and evaluate the outputs of F1 and F2 as functions of the four inputs. 5mks
 ii. List the truth table with 16 binary combinations of the four inputs variables. 4mks
 The truth table should also include values for T1 to T4 as well as F1 and F2.

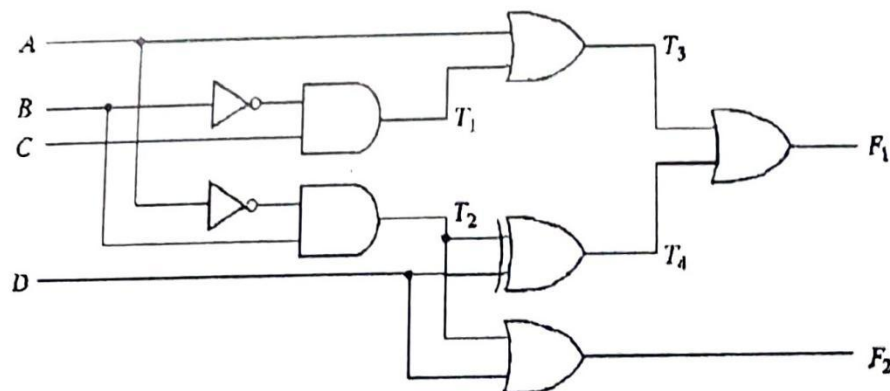


Figure 1

- b. Compare Von Neumann Architecture and Harvard Architecture. Include diagram for each. 4mk
 c. Mention any 3 clear differences between RISC and CISC. Give two examples of processors in each category. 3mks
 d. Describe End-Around Carry in 1's Complement? Illustrate this using 153 – 142 in a 10-bit register. 4mks